

[XR Podcast Explorer]

Testing plan for interactive prototype 1/2b

XR Podcast Explorer reimagines a traditional podcast browsing application as a spatial XR experience. Instead of only using conventional menus, users can explore podcast topics through spatial doors, browse episodes and chapters in virtual spaces, and physically place a heart to favourite content. The project uses XR to explore whether spatial navigation and embodied interactions can make podcast browsing more engaging and intuitive.

Testing Objective

From my above concept, I have identified an unvalidated assumption that first-time users will understand the overall spatial navigation and interaction flow without detailed instructions.

This includes whether users can understand how to enter a topic through a door, browse and select episodes and chapters, enter a selected chapter through a door, favourite content using a physical heart, and navigate between the main spaces.

This test aims to discover which interactions users can complete independently and where they experience confusion, hesitation, incorrect actions, or require assistance. The results will help identify which parts of the interaction flow need to be improved in the next prototype iteration.

Testing Methodologies

This testing plan will use **task-based usability testing, observation, think-aloud, and short post-test questions** to evaluate a digital prototype made in Unity.

Participants will be given goals to complete without being told exactly which objects or controls to use. During the tasks, participants will be encouraged to think aloud while I observe their actions, errors, hesitation, and whether they can complete each task independently.

After completing the tasks, I will ask several short questions to understand which interactions were intuitive or confusing and collect suggestions for improvement.

Prototype description/requirements

The prototype was designed to provide enough interaction to test whether users can understand the overall navigation and interaction flow of the XR Podcast Explorer.

It is a horizontal Unity prototype containing four main spaces:

Podcast Hub: Users can explore different podcast topics represented by doors and enter a selected topic.

Episode Room: Users can use Previous and Next controls to browse the available episodes, view changing episode information, and select an episode.

Chapter Portal: Users can browse available chapters using the Navigation Panel, select a chapter, and enter the selected chapter through the Current Chapter Door. The Navigation Panel also contains Home and Episode navigation.

Podcast Experience: Users can experience sample podcast content and favourite it by taking a heart from the Heart Shelf and placing it onto the Favourite Area. A Navigation button allows users to access the Navigation Panel again.

Teleportation is used to move between the main virtual spaces so the prototype can focus on testing the overall interaction flow.

Data collection method

Data will be collected through **direct observation, think-aloud comments, task performance measures, and a short post-test interview.**

A structured testing sheet will be used for each participant. For each task, the following measures will be recorded:

- **Task success:** whether the participant completes the intended task successfully.
- **Completion time:** the approximate time taken from receiving the task to successful completion.
- **Errors:** the number of incorrect interactions or navigation choices made during the task.
- **Hesitation:** noticeable uncertainty or pauses when deciding what to do next.
- **Assistance required:** whether a moderator hint is required for the participant to continue.

- **Observations and think-aloud comments:** notable behaviours, expectations, confusion, and comments expressed during interaction.

Additional task-specific observations will be recorded, including whether participants browse all Episodes and Chapters, understand the relationship between Chapter selection and the Current Chapter Door, recognise the Heart-based favourite interaction, and locate the Navigation controls without assistance.

Following completion of all tasks, a short semi-structured interview will collect qualitative feedback about interaction clarity, spatial navigation, the favourite interaction, and possible improvements.

Participants will be identified anonymously as P1, P2, P3, etc. Results will be compared across participants to identify repeated interaction problems and actionable opportunities for the next design iteration.

Task	Interaction / Function	Assumption being tested	Method	Key Metrics
T1 Topic Selection	Topic Door	Users recognise doors as podcast topics and understand how to enter one	Task-based testing + Think-aloud	Success, time, errors, hesitation
T2 Episode Browsing	Episode controls	Users can discover the browsing controls and understand changing episode information	Task-based testing + Observation	Browse-all success, time, errors, assistance
T3	Navigation	Users understand that chapter selection	Task-based	Success, hesitation,

Task	Interaction / Function	Assumption being tested	Method	Key Metrics
Chapter Navigation	Panel + Chapter Door	changes the Current Chapter Door	testing + Think-aloud	errors, assistance
T4 Favourite	Heart + Favourite Area	Users understand the physical Heart interaction without conventional button instructions	Task-based testing + Think-aloud	Success, time, failed attempts, assistance
T5 Global Navigation	Navigation Panel	Users understand how to return to Episode Room and Podcast Hub	Task-based testing + Observation	Success, navigation errors, assistance

Testing Setup

The testing session requires:

- A computer running the completed Unity horizontal prototype.
- The prototype opened at the **Podcast Hub**, which is the starting point of the experience.
- All interactive doors reset to their closed state.
- Episode and Chapter selections reset to their initial states.
- The Navigation Panel reset to its intended initial visibility state.
- The Heart returned to the Heart Shelf.
- A timer for recording approximate task completion time.
- One testing results sheet for each participant.

Before testing begins, the prototype should be checked by completing the main interaction flow once to confirm that Topic selection, Episode browsing, Chapter navigation, Podcast Experience, Favourite interaction, and Navigation are functioning correctly.

Participants should complete the test individually. The moderator should remain beside the participant to present each task, record observations, and provide assistance only when necessary.

No interaction should be demonstrated before the test. Participants should be encouraged to think aloud by describing what they are looking for, what they expect an interaction to do, and anything they find confusing.

The prototype should be fully reset before the next participant begins.

Testing process: (also considering the schedule/time)

The testing session is designed to take approximately **five minutes per participant**, consistent with the recommended IP1 testing duration.

1. Participant Introduction — approximately 30 seconds

Moderator script:

“Thank you for participating in this prototype test. This is an XR podcast exploration experience. The purpose of this session is to evaluate the design rather than your performance, so there are no right or wrong actions.

You will be given a series of tasks one at a time. Please try to complete each task without instructions about which controls to use. While interacting with the prototype, please describe what you are thinking, what you expect to happen, and anything you find confusing. If you become completely stuck, assistance can be provided so that the remaining tasks can continue.”

No demonstration of the prototype interactions is provided.

2. Task 1 — Topic Selection — approximately 30 seconds

Task instruction:

“Please explore the Podcast Hub and enter the AI podcast topic.”

Interaction being evaluated:

Understanding of spatial Topic Doors as the primary method of topic selection.

Observation criteria:

- Recognition of the doors as selectable podcast topics.
- Selection of the correct topic.
- Successful interaction with the Topic Door.
- Incorrect interactions or hesitation before entering the Episode Room.

Completion condition:

The participant successfully reaches the Episode Room.

3. Task 2 — Episode Browsing and Selection — approximately 45 seconds

Task instruction:

“Please browse through all of the available episodes. After viewing all of them, select the episode you would most like to explore.”

Interaction being evaluated:

Understanding of the Episode browsing controls and Episode selection process.

Observation criteria:

- Discovery of the Episode browsing controls without instruction.
- Successful browsing through **all available Episodes**.
- Recognition that the Episode title and information change during browsing.
- Successful selection of an Episode.
- Errors or hesitation when browsing or selecting.

The names of specific controls such as Previous and Next should not be provided during the task.

Completion condition:

All Episodes have been viewed and the participant successfully selects an Episode and reaches the Chapter Portal.

4. Task 3 — Chapter Browsing and Selection — approximately 45 seconds

Task instruction:

“Please browse through all of the available chapters. After viewing all of them, select the Interview chapter and enter it.”

Interaction being evaluated:

Understanding of Chapter navigation and the relationship between Chapter selection and the Current Chapter Door.

Observation criteria:

- Recognition and use of the Chapter Navigation Panel.
- Browsing through **all available Chapters**.
- Successful selection of Interview.
- Recognition that the Current Chapter Door represents the selected Chapter.
- Successful interaction with the Current Chapter Door.
- Errors or hesitation between selecting and entering the Chapter.

The relationship between the Chapter Navigation Panel and Current Chapter Door should not be explained during the task.

Completion condition:

The participant successfully enters Interview and reaches the Podcast Experience.

5. Task 4 — Favourite Interaction — approximately 45 seconds

Task instruction:

“Please save this podcast content as one of your favourites.”

Interaction being evaluated:

Whether the physical Heart interaction communicates the favourite function without relying on a conventional Favourite button.

Observation criteria:

- Recognition of the Heart Shelf.
- Recognition that the Heart is interactive.
- Understanding of the relationship between the Heart and Favourite Area.
- Successful movement and placement of the Heart onto the Favourite Area.
- Initial incorrect actions, such as attempting to click the Favourite Area.
- Hesitation or assistance required.

The terms **Heart**, **Heart Shelf**, **Favourite Area**, **drag**, and **place** should not be included in the task instruction.

Completion condition:

The Heart is successfully attached to the Favourite Area.

6. Task 5 — Global Navigation — approximately 45 seconds

Task instruction:

“Without restarting the experience, return to the Episode Room. After that, return to the Podcast Hub.”

Interaction being evaluated:

Understanding of the Navigation control and the Episode/Home navigation options.

Observation criteria:

- Discovery of the Navigation button without instruction.
- Understanding that Navigation reveals the Navigation Panel.
- Correct use of Episode to return to the Episode Room.
- Correct use of Home to return to the Podcast Hub.
- Incorrect navigation selections or hesitation.

The Navigation, Episode, and Home controls should not be identified by the moderator during the task.

Completion condition:

The participant successfully returns to the Podcast Hub.

7. Assistance Protocol

Moderator assistance should be minimised to avoid influencing the participant's natural interaction with the prototype.

If no progress is made for approximately **15 seconds**, one neutral prompt may be provided:

“Please take another look at the available objects and controls.”

If further assistance is required to continue the test, the interaction may be explained sufficiently to allow progression to the next task.

Any moderator assistance must be recorded as:

Assistance Required = Yes

The point at which assistance was required should also be recorded in the observation notes.

8. Post-Test Interview — approximately 1 minute

Following completion of the five tasks, the same questions are presented to every participant:

1. **Which interaction was the easiest to understand, and why?**
2. **Was there any point where you were unsure about what to do next?**
3. **What did you think the Heart and Favourite Area were for when you first saw them?**
4. **Did the movement between Topic, Episode, Chapter and Podcast Experience feel clear? Why or why not?**
5. **If you could change one thing about the experience, what would you change?**

Short responses and notable participant comments are recorded in the testing results sheet.

The prototype is then reset to its initial state before the next participant begins.